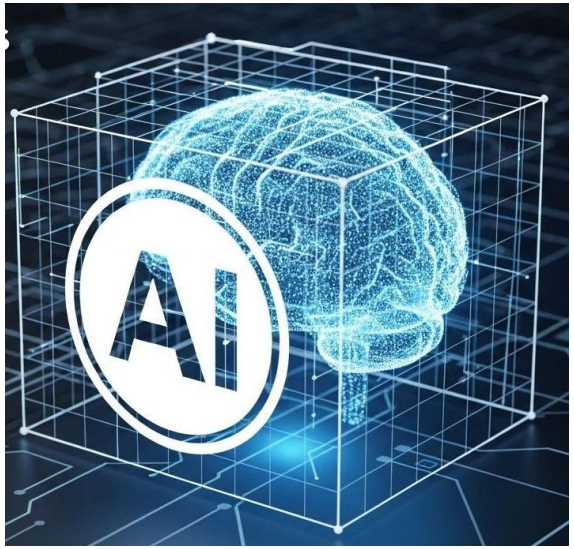


ALY 3510 · Fall 2026

Lecture 8: Robotics & Embodied AI - Appendix

Intelligence that acts in the physical world

Appendix: Slides, Video for Self-Learning

VIDEO: Particle Filters for Localization (2:10)



What to Watch For:

- Particles represent possible robot positions (hypotheses)
- Particles converge as sensor data confirms location
- More particles = better accuracy but higher computation

Autonomous Flight Using Waypoints (2:31)



Autonomous flight using **waypoints**

Military AI & Ethical Considerations

Military Robotics Applications

- Reconnaissance & surveillance drones
- Bomb disposal robots
- Autonomous logistics vehicles
- Search and rescue operations
- Perimeter security systems

Ethical Concerns

- Lethal autonomous weapons
- Accountability for AI decisions
- Lowered threshold for conflict
- Arms race dynamics

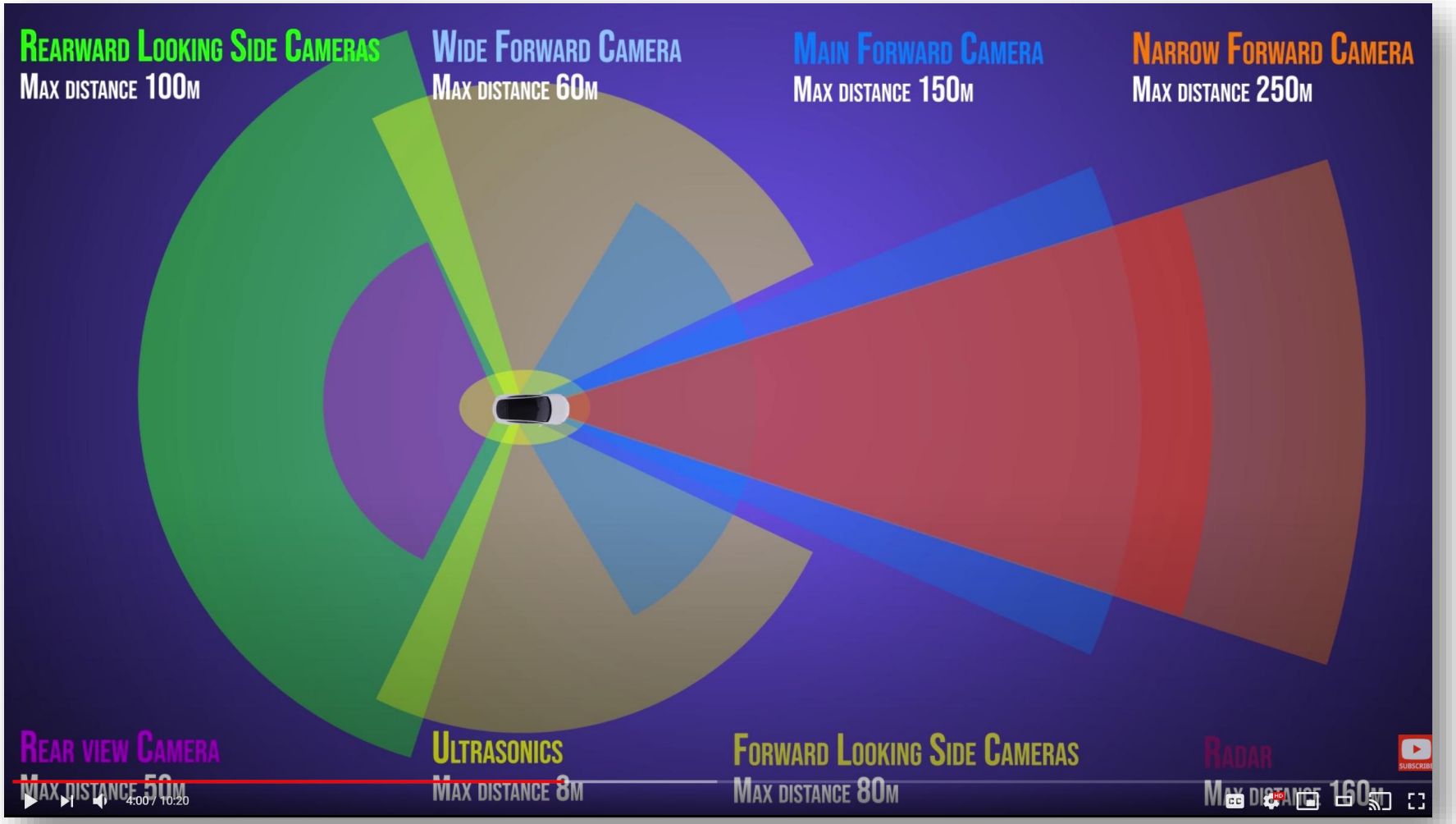
Key Question: Should machines make life-or-death decisions without human oversight?

Emerging Principles

- Human-in-the-loop: Humans approve critical decisions
- Meaningful human control
- International treaties & regulations
- Industry self-regulation efforts



Sensors: Cameras, Lidar, Radar, Ultrasonic



LIDAR & Point Cloud Processing

LIDAR (Light Detection and Ranging): Uses laser pulses to measure distances and create 3D maps of the environment

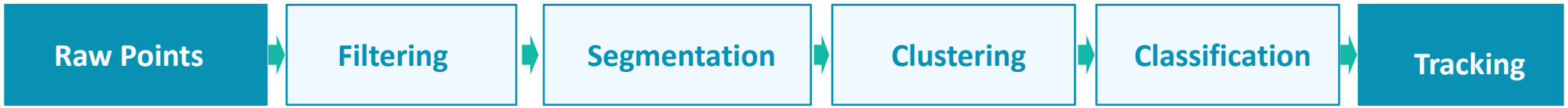
How LIDAR Works

1. Emit a laser pulse
2. Light reflects off objects
3. Measure return time
4. Calculate distance (speed of light)
5. Repeat 100,000+ times/second

Point Cloud Output

- Millions of 3D points (x, y, z)
- Real-time 360° environment model
- Detects objects, surfaces, edges
- Centimeter-level accuracy

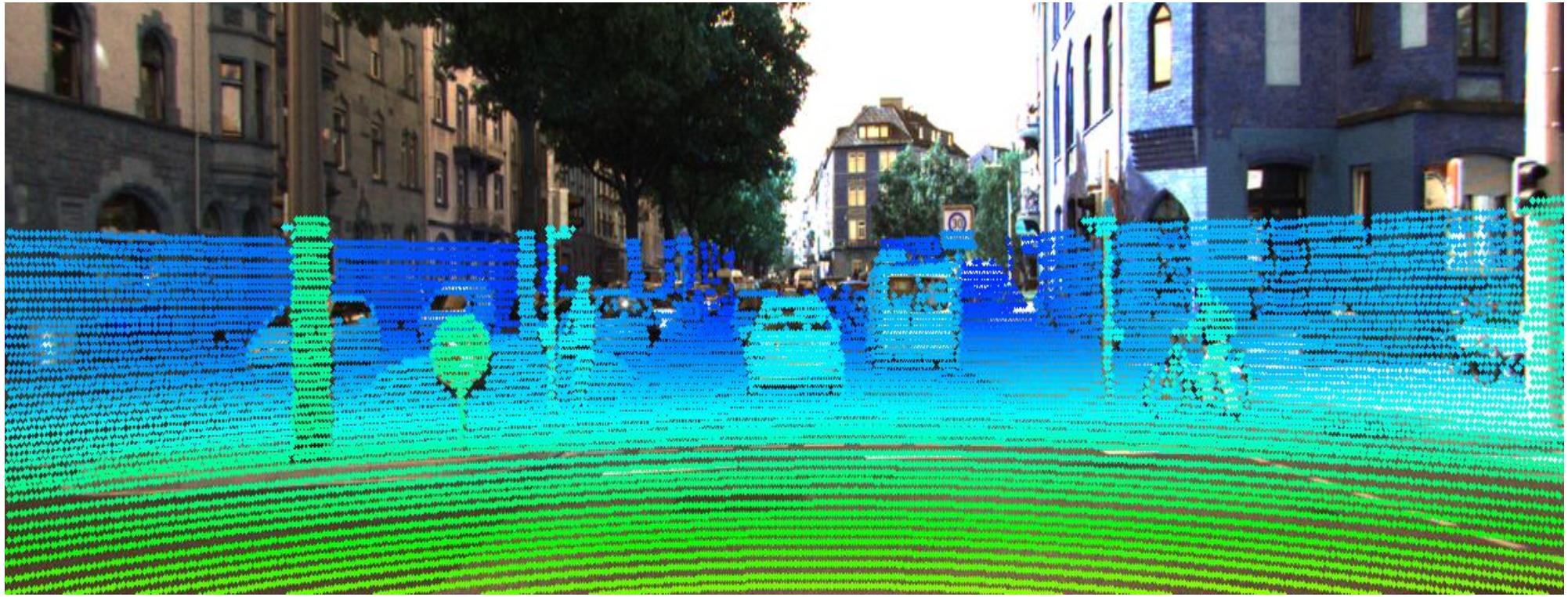
Point Cloud Processing Pipeline



Result: Real-time understanding of vehicles, pedestrians, cyclists, and obstacles

Autonomous Cars – LIDAR Technology

LIDAR - Light Detection And Ranging: (point cloud) + Video



Urban Driving Challenges

Why city driving remains the hardest problem for autonomous vehicles

Unpredictable Agents

- Pedestrians jaywalking
- Cyclists weaving
- Other erratic drivers

Complex Intersections

- Unprotected left turns
- 4-way stops
- No lane markings

Edge Cases

- Construction zones
- Emergency vehicles
- Unusual objects

Weather & Lighting

- Rain, snow, fog
- Sun glare
- Night conditions

Infrastructure Variety

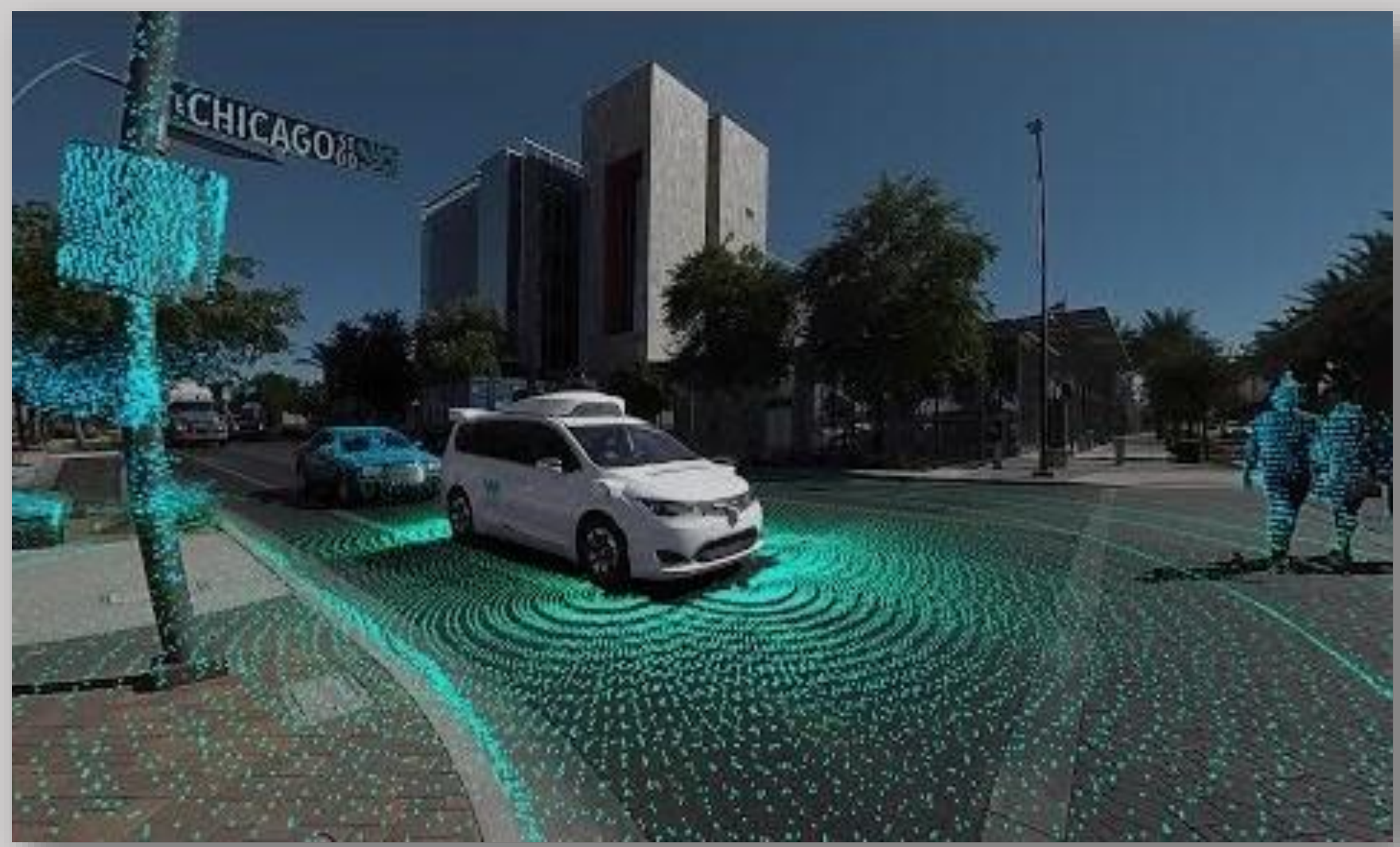
- Faded lane lines
- Missing signs
- Different regions

Social Navigation

- Eye contact cues
- Hand gestures
- Cultural norms

The "long tail" problem: Rare events are infinite in variety. Each new rdge case requires new training data or explicit handling.

VIDEO: Urban Driving by Waymo (3:36)



Degrees of Freedom (DoF)

Degrees of Freedom: The number of independent ways a robot can move. More DoF = more flexibility but more complexity.

Simple Gripper

1 DoF

Open/close only

Basic pick & place

Industrial Arm

6 DoF

Reach any position and orientation/rotation

Welding, assembly

Human Arm + Hand

27+ DoF

Incredible dexterity

Fine manipulation

Why 6 DoF Matters

6 DoF is the minimum needed to reach any point in 3D space with any orientation.

- 3 for position (x, y, z) + 3 for orientation/rotation (roll, pitch, yaw).
- More DoF adds redundancy and flexibility.

End Effectors: Robot "Hands"

The tool at the end of a robot arm that interacts with the world

Parallel Jaw

Two fingers, simple & reliable

Use: Boxes, regular shapes

Vacuum Suction

Air pressure gripping

Use: Flat surfaces, sheets

Magnetic

Electromagnet attachment

Use: Metal parts, automotive

Soft Gripper

Compliant, adaptable fingers

Use: Delicate, irregular items

Multi-Finger Hand

Dexterous manipulation

Use: Complex assembly tasks

Tool Changers

Swap end effectors automatically

Use: Multi-task flexibility

Selection criteria: Object shape, weight, surface texture, fragility, speed requirements, and cost

Dexterous Robot Hands

Approaching human-level manipulation capability

Why Build Human-Like Hands?

- World designed for human hands
- Tool use without modification
- In-hand manipulation (rotate, adjust)
- Natural human-robot collaboration

Technical Challenges

- 20+ DoF to control simultaneously
- Compact, powerful actuators
- Rich tactile sensing
- Real-time control algorithms

Notable Dexterous Hands

Shadow Hand	Allegro Hand	LEAP Hand	Tesla Bot Hand
Shadow Robot	Wonik Robotics	Carnegie Mellon	Tesla
24 DoF, air muscles	16 DoF, research standard	Low-cost, 3D printed	22 DoF, humanoid integration

Trend: AI + dexterous hands enabling unprecedented manipulation capabilities

Bipedal Locomotion: Walking Like Humans

The engineering challenge of two-legged walking

Walking is "controlled falling" — constantly losing and catching balance with every step

Key Concepts

- Zero Moment Point (ZMP)
  Keep center of pressure within foot
- Dynamic Balance
  Constant adjustments, not static sta
- Gait Planning
  Foot placement, stride, timing patte
- Terrain Adaptation
  Stairs, slopes, uneven surfaces

Progress: From Lab to Real World

2000



https://www.sanctuary.ai/blog/sanctuary-ai-unveils-phoenix-a-humanoid-general-purpose-robot-designed-for-work

The Quest for General-Purpose Robots

From specialized machines to versatile assistants

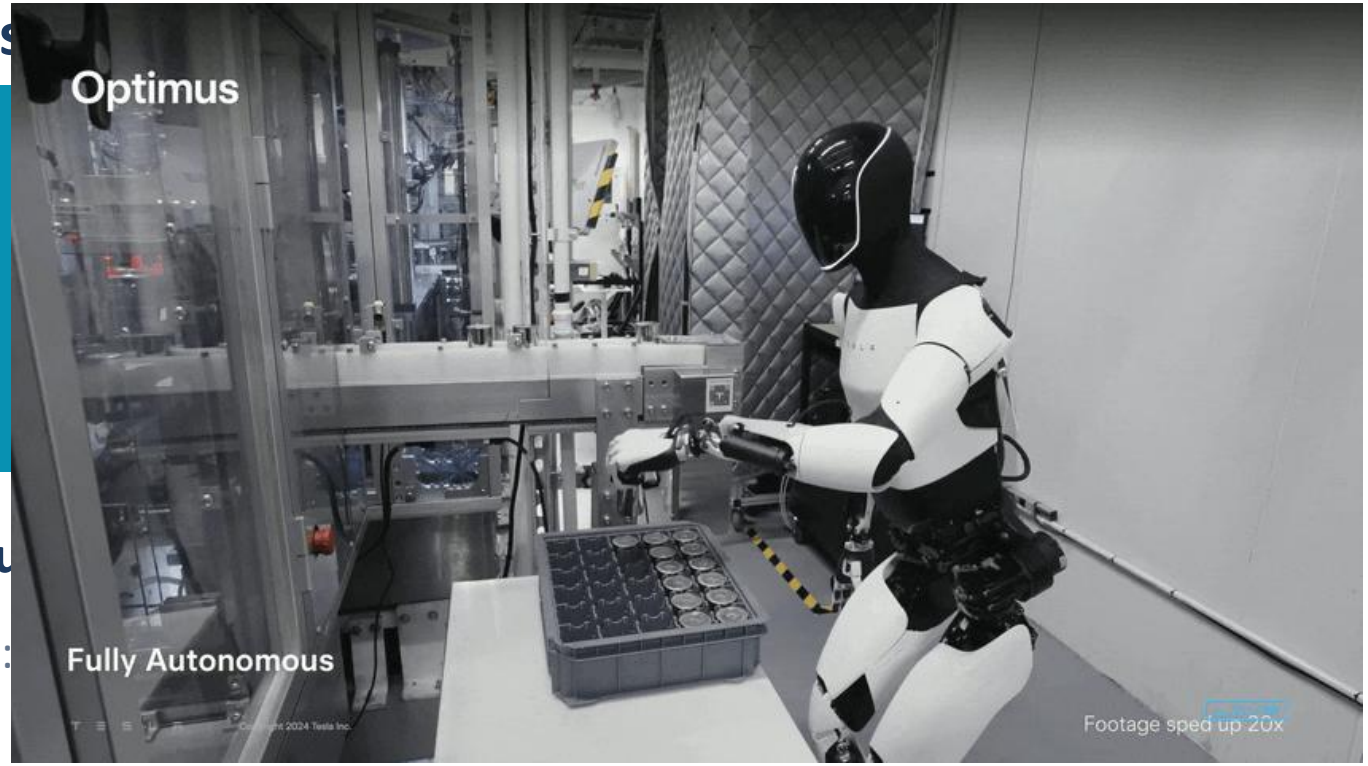
The Specialization Spectrum

Single Task	Domain Specific	Multi-Task	General Purpose
Welding arm	*Warehouse bot*	*Cobot*	*Humanoid?*

What Makes a Robot "General Purpos

Required Capabilities

- Learn new tasks quickly (few-shot)
- Transfer skills between domains
- Handle unstructured environments
- Recover from errors autonomously



When will we have tru

Optimistic: 5-10 years • Realistic: